z/OS Systems Programming Essentials

INTRODUCTION AND CLASS LOGISTICS


TEKsystems Global Services
WORKFORCE DEVELOPMENT

LOGISTICS

Class Hours:

- Instructor will provide class start and end times.
- There will be regular breaks in class.

Telecommunication:

- Turn off or set electronic devices to silent (not vibrate)
- Reading or attending to devices can be distracting to other students
- Try to delay until breaks or after class

Miscellaneous:

- Courseware
- Bathroom
- Fire drills

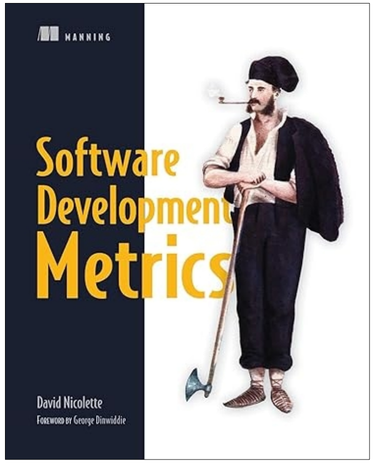
DAVE NICOLETTE

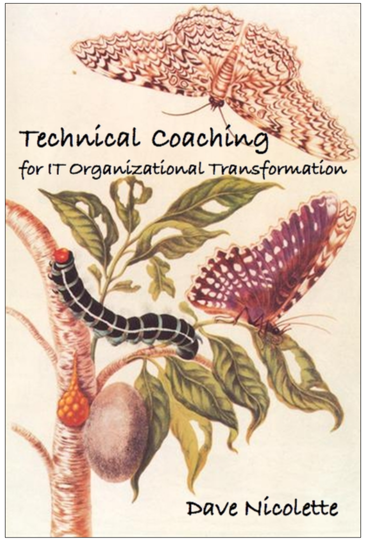
Software Developer

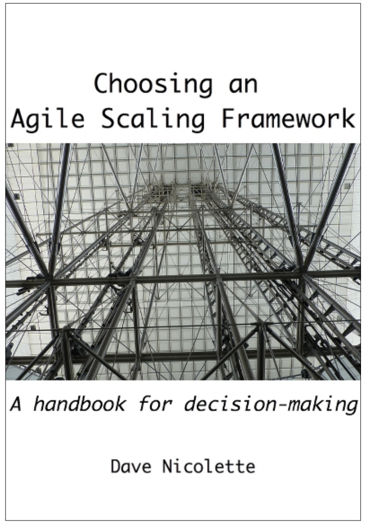
Technical Coach

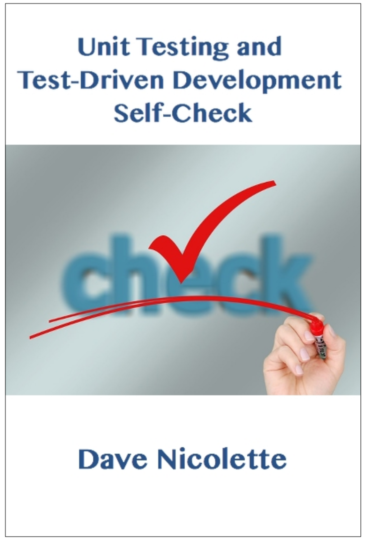
Speaker / Author

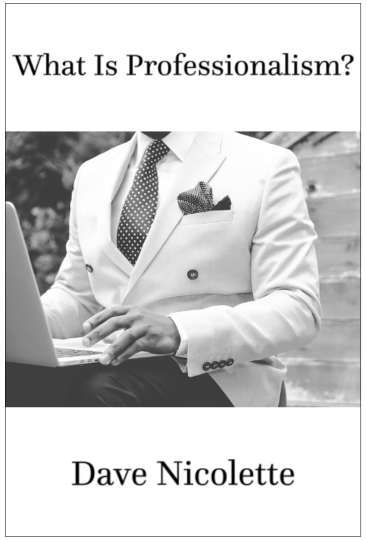
@davenicolette@mastodon.social











INTRODUCE YOURSELF

Time to introduce yourself:

- Name
- What is your role in the organization
- Indicate software-related or IT experience

LABS

A significant portion of the class time will be spent in hands-on work.

We will use contemporary collaborative methods such as Ensemble Programming and Pair Programming.

Everyone is expected to participate and to ask for and offer help. It is not a competition.

Bootcamp Overview

SCOPE AND OBJECTIVES

1

SCOPE AND OBJECTIVES OF THE BOOTCAMP

The six-week z/OS bootcamp aims to provide you with basic knowledge and skills to begin working effectively with the IBM z/OS operating system.

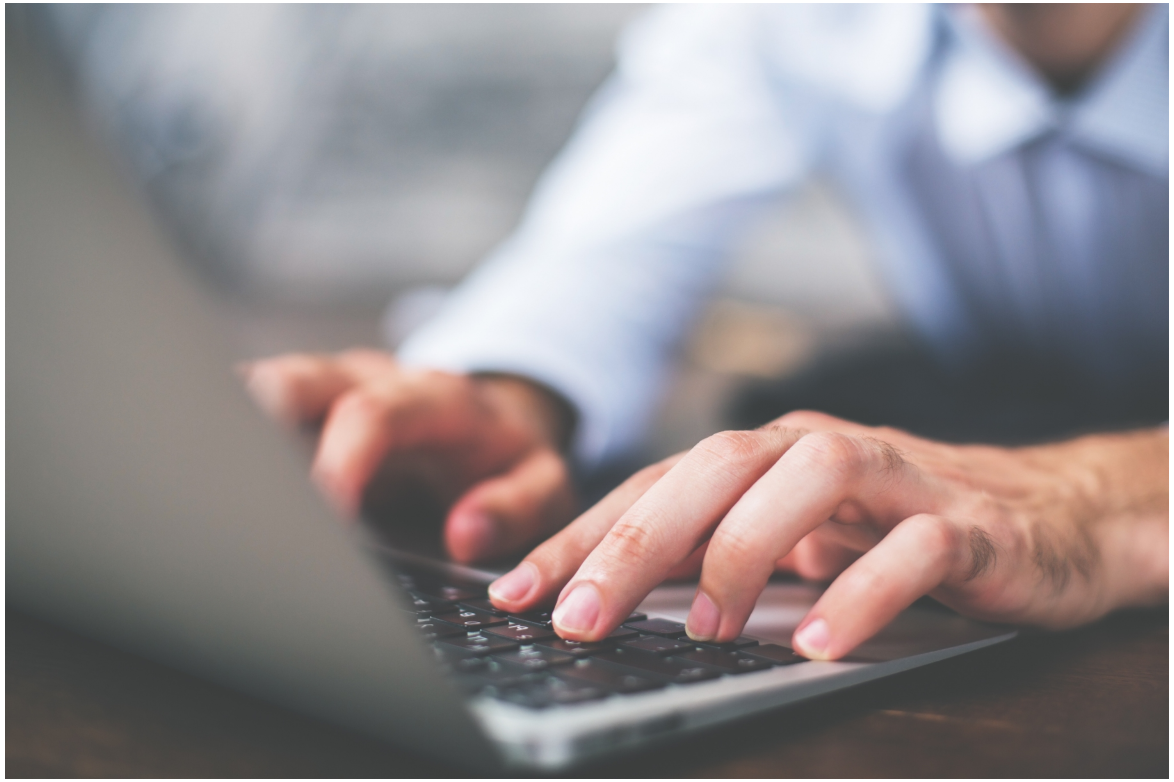
The key skill areas to cover are:

- Some z/OS basics
- TSO and ISPF (Time-Sharing Option)
- JCL (Job Control Language)
- Software installation & management using SMP/E and z/OSMF
- Ansible for z/OS automation
- Infrastructure automation tools

Getting Our Bearings

COURSE REPOSITORY, TERMINAL EMULATOR, CONNECTING TO Z/OS

RESOURCES FOR HANDS-ON WORK



We'll take a hands-on approach throughout the training. To enable that, you'll need access to three resources:

- The repository containing starter code, data files, and sample code.
- A 3270 terminal emulator installed locally on your system or accessible to you on your network.
- A user ID and password for the z/OS lab system
- A user ID and password for the Ansible control node

The expectation is that you have received the URL for the repository and credentials for the lab system and you have access to a 3270 emulator. Best case, you have tried these things already to ensure they are functional.

Let's make sure you have everything in place and you can access what you will need to complete the work.

JUNE 26 USERIDS AND INITIAL PASSWORDS

Kamal Kharel	GAMA11
Chavan Srushti	GAMA12
Jeremy Griffith	GAMA13
Michael DiGiovanna	GAMA14
Miles Newland	GAMA15
Ray Thomas	GAMA16
John DeLuce	GAMA17
Alex Kanji	GAMA18
Lucas Franklin	GAMA19
Andrew Cook	GAMA20

COURSE REPOSITORY

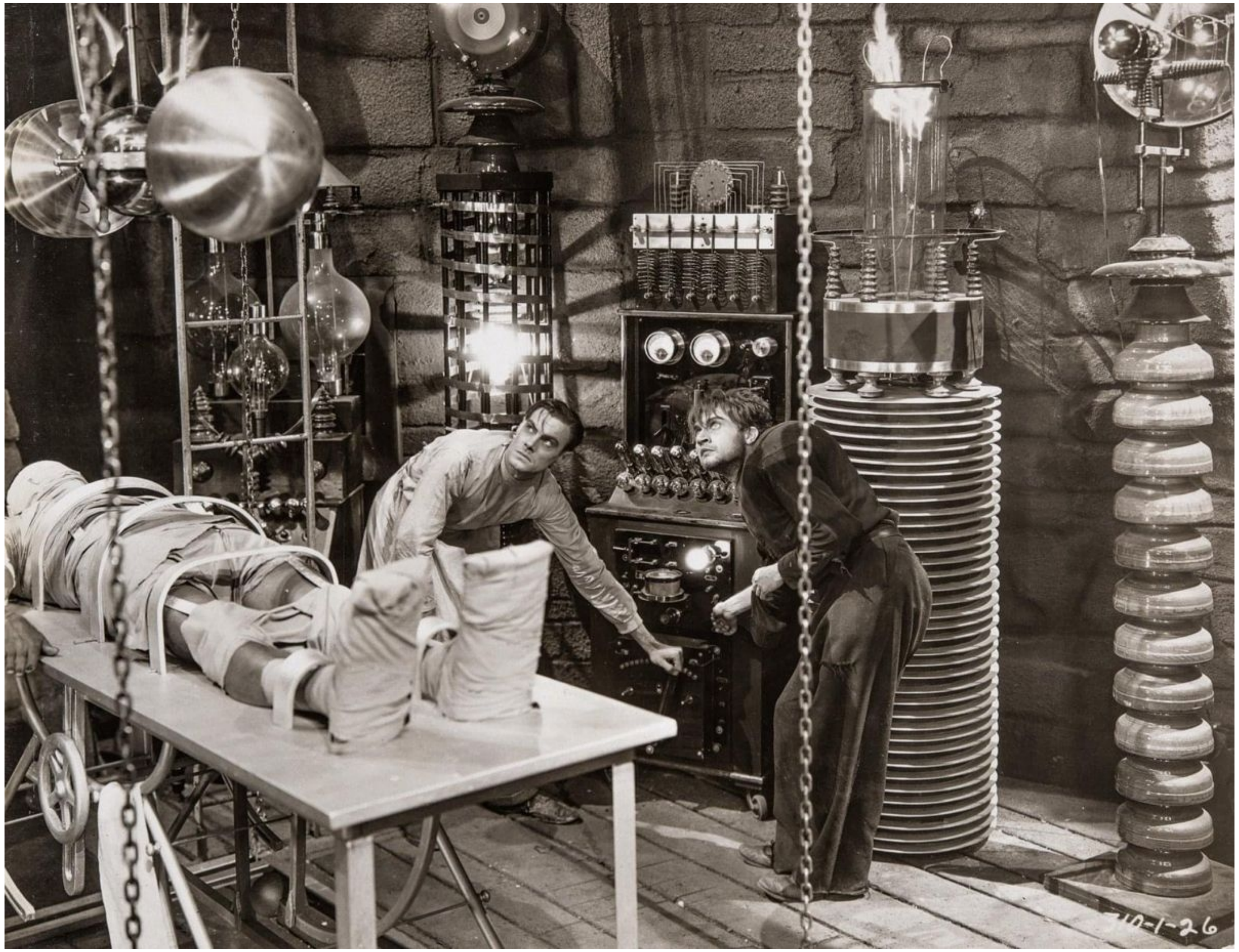
The course repository contains instructions, data files, and starter code for lab work and the capstone project, as well as complete sample code for reference. You should have been given the URL in advance.

You can clone the repo or download selected files from it. To push your work, you must be a committer.

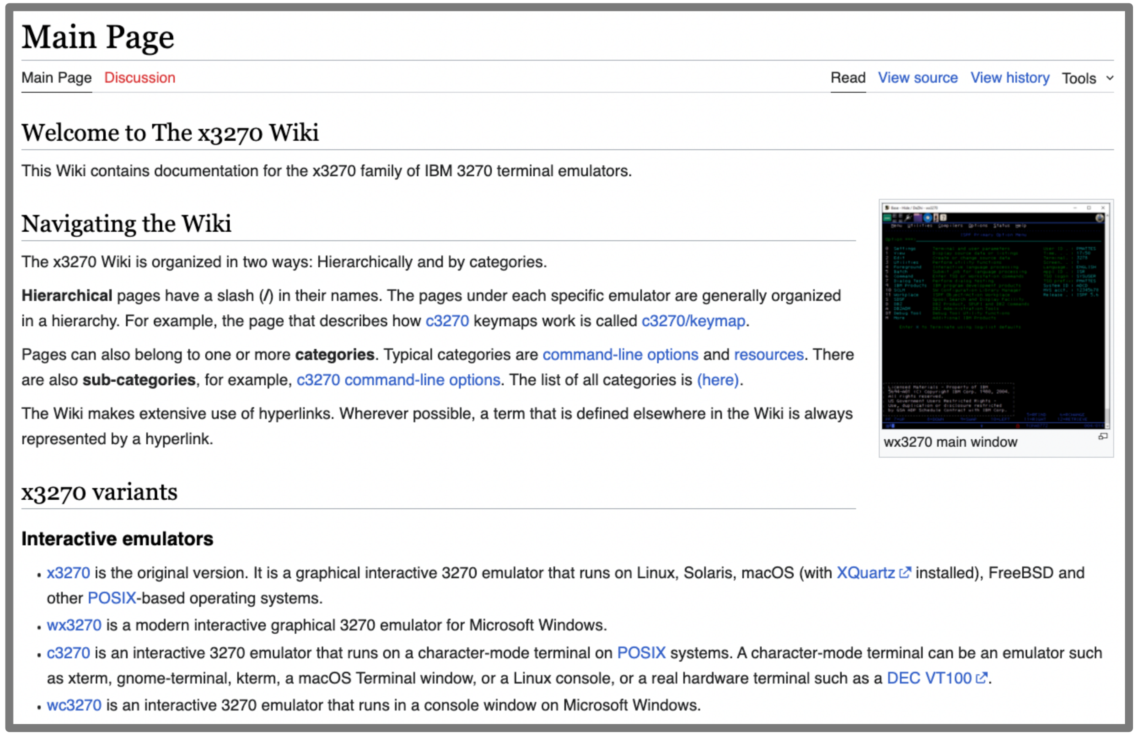
If you have not already set this up, then we'll do it now. You need a Github ID. Your instructor can then send you an invite within Github, to which you must reply in order to become a committer.

After that, you can push your work to the repo for future reference and to share with colleagues.

LAB Z/OS 1: CLONE THE COURSE REPOSITORY



TERMINAL EMULATOR


Main Page

Main Page Discussion Read View source View history Tools

Welcome to The x3270 Wiki

This Wiki contains documentation for the x3270 family of IBM 3270 terminal emulators.

Navigating the Wiki

The x3270 Wiki is organized in two ways: Hierarchically and by categories.

Hierarchical pages have a slash (/) in their names. The pages under each specific emulator are generally organized in a hierarchy. For example, the page that describes how c3270 keymaps work is called c3270/keymap.

Pages can also belong to one or more **categories**. Typical categories are command-line options and resources. There are also **sub-categories**, for example, c3270 command-line options. The list of all categories is (here).

The Wiki makes extensive use of hyperlinks. Wherever possible, a term that is defined elsewhere in the Wiki is always represented by a hyperlink.

wx3270 main window

x3270 variants

Interactive emulators

- x3270 is the original version. It is a graphical interactive 3270 emulator that runs on Linux, Solaris, macOS (with XQuartz installed), FreeBSD and other POSIX-based operating systems.
- wx3270 is a modern interactive graphical 3270 emulator for Microsoft Windows.
- c3270 is an interactive 3270 emulator that runs on a character-mode terminal on POSIX systems. A character-mode terminal can be an emulator such as xterm, gnome-terminal, kterm, a macOS Terminal window, or a Linux console, or a real hardware terminal such as a DEC VT100.
- wc3270 is an interactive 3270 emulator that runs in a console window on Microsoft Windows.

For historical reasons, the basic way to access a z/OS system is through an IBM 3270 terminal. Physical terminals are no longer used. People use 3270 emulation software to interact with z/OS.

If you have not already set up a 3270 terminal emulator, please take a few minutes to install one now. You will not be able to complete any lab assignments or the capstone project without one.

Your company may have an emulator or a third-party graphical front-end (e.g., Zowe) they want you to use. Otherwise, the most popular free 3270 emulator is x3270. See the wiki for a version that runs on your local system:

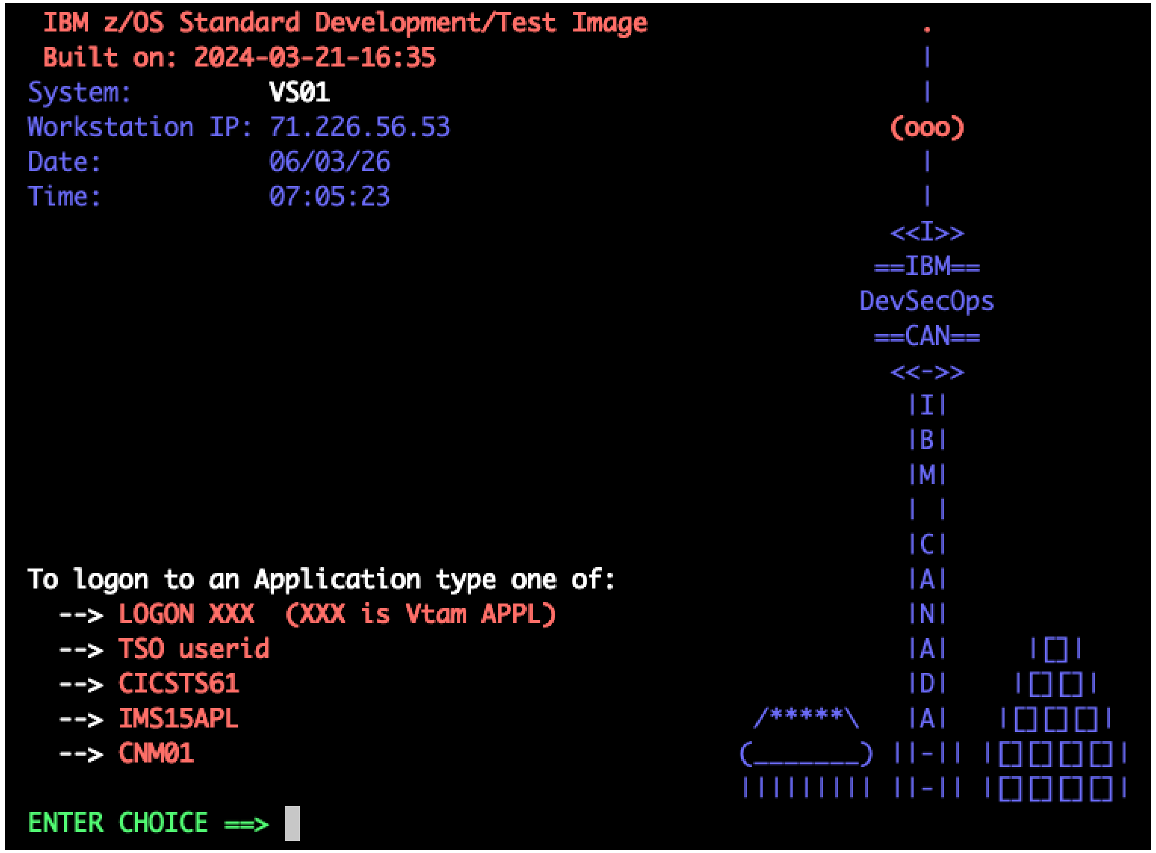
`https://x3270.miraheze.org/wiki/Main_Page`

ACCESSING THE LAB Z/OS ENVIRONMENT

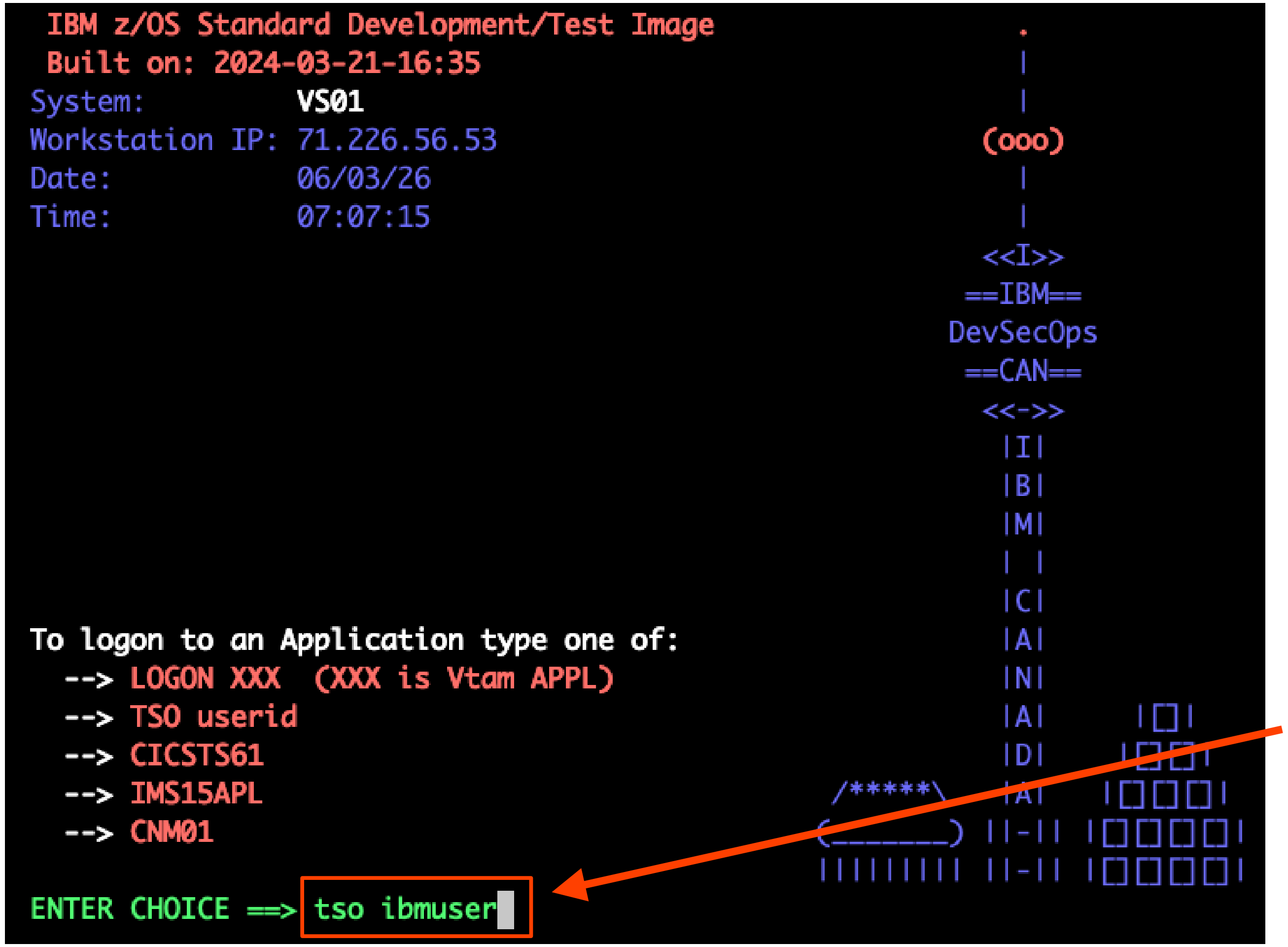
The URL, port, user ID, and password should have been provided to you beforehand.

Try to connect now and we will deal with any issues.

When you connect, the display will resemble the example at right.

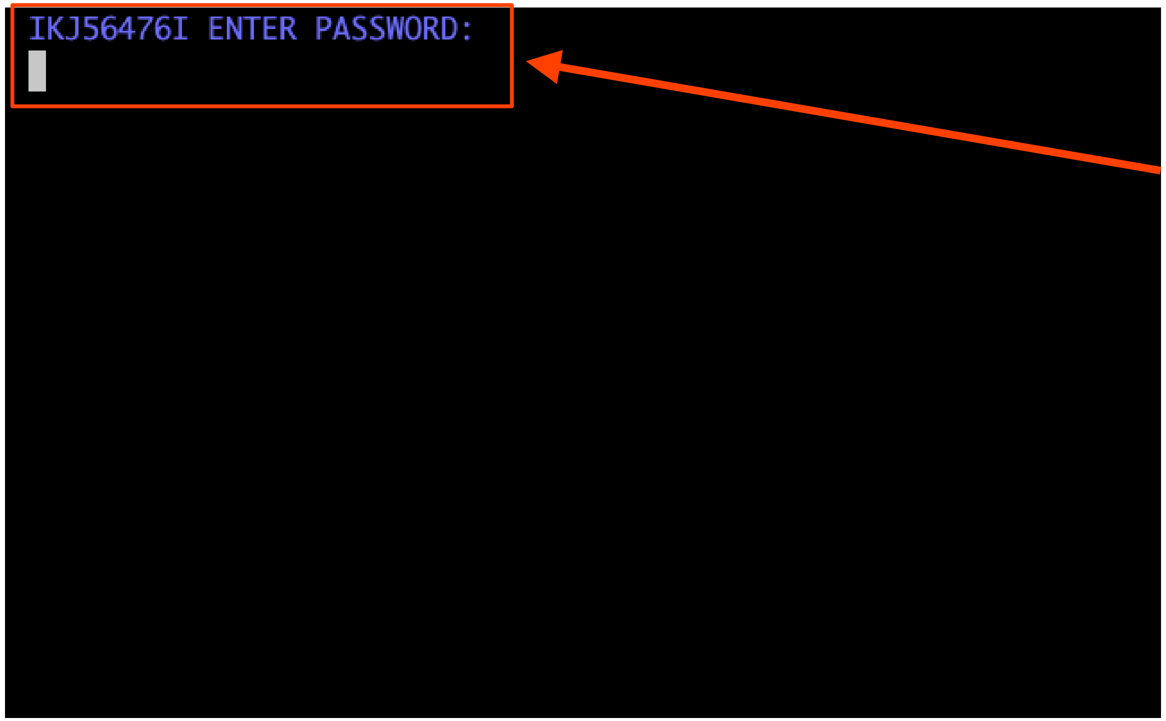


ACCESSING THE LAB ENVIRONMENT



At the Command prompt, type **tso** followed by your userid and press Enter.

ACCESSING THE LAB Z/OS ENVIRONMENT



The lab system prompts you for your password. It may be difficult to read because of the default colors of the terminal emulator.

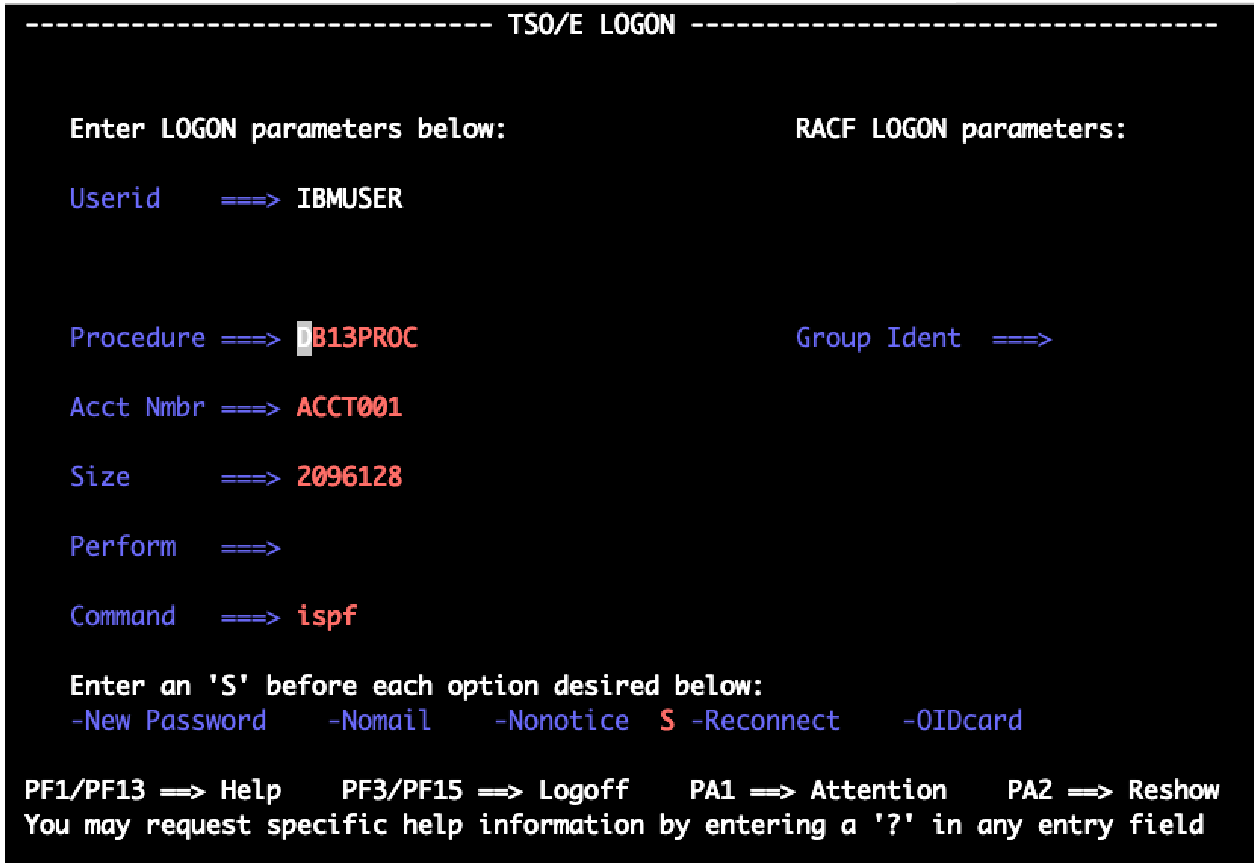
The system at your workplace may look different – this is configurable, and most organizations present their own sign-on form, or no form.

What you see here is the bare-bones default password prompt.

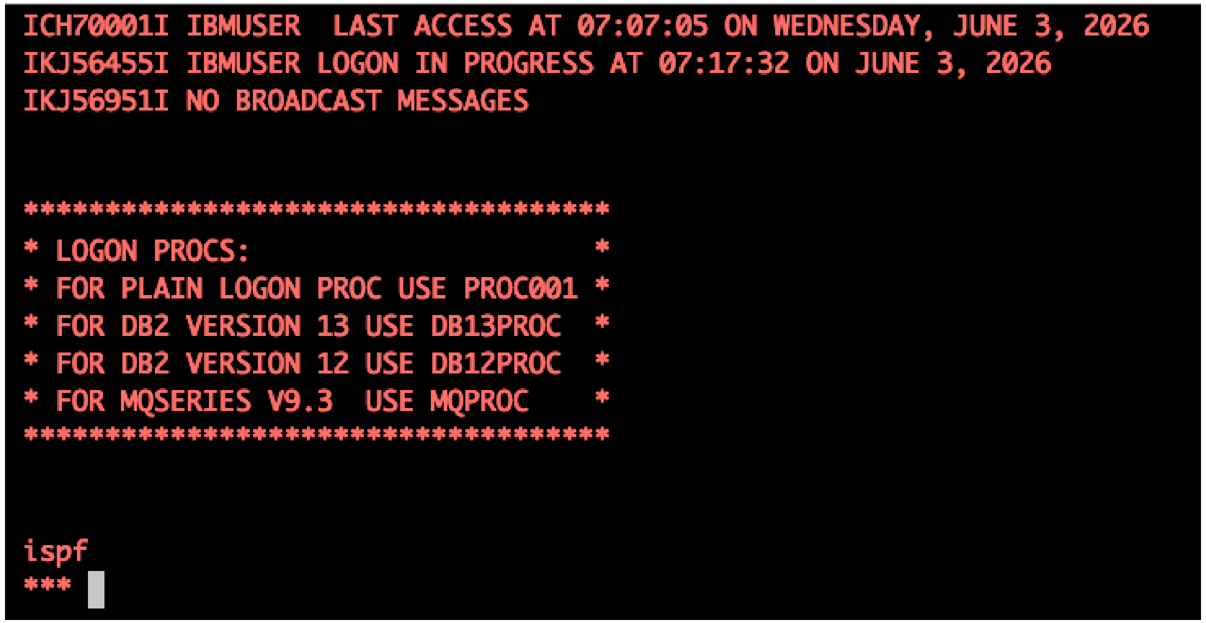
ACCESSING THE LAB Z/OS ENVIRONMENT

Normally, the next thing you will see is the TSO/E Logon panel, where you can specify a startup script to set up your ISPF session. You do not need to change anything on this screen. Press Enter to dismiss it.

The system may be configured to require you to change your password the first time you log on. In that case, you will see the password change panel before the TSO/E Logon panel.



ACCESSING THE LAB Z/OS ENVIRONMENT



After the TSO/E Logon panel, TSO displays any broadcast messages and any notifications for your userid, plus some comments advising you about available logon procs.

Near the bottom of the screenshot you see "ispf". That's the command to start ISPF from the TSO prompt.

The three asterisks at the bottom are telling you that you can dismiss these messages by pressing Enter.

ACCESSING THE LAB Z/OS ENVIRONMENT

ISPF (the Interactive System Productivity Facility) is a TSO application that provides a relatively user-friendly interface (compared to the bare TSO command line) to TSO commands, a text editor, and access to various system utilities. This is the main user interface for application programmers and system programmers.

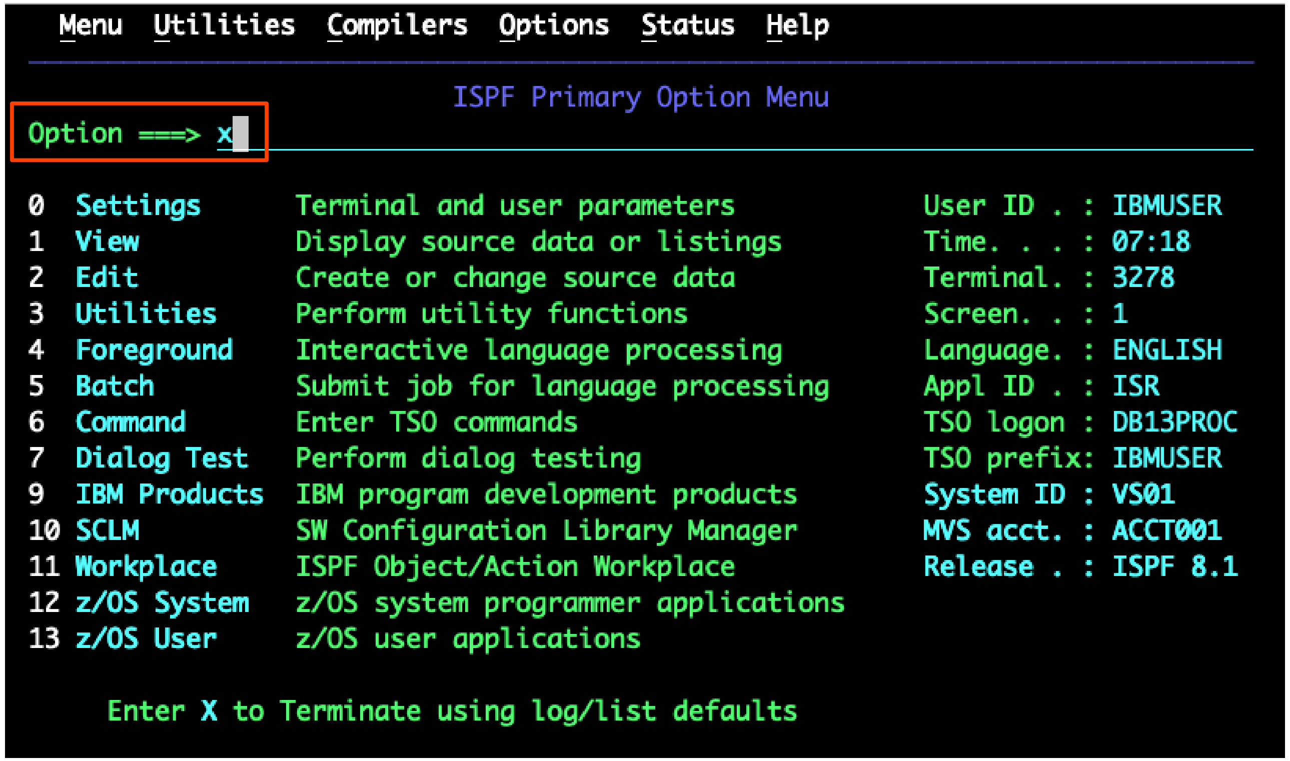
It's a menu-driven system that presents formatted panels to help you enter the correct parameters for system commands and tools.

```
  Menu  Utilities  Compilers  Options  Status  Help
------------------------------------------------------------------------------
                           ISPF Primary Option Menu
Option ===>

0  Settings      Terminal and user parameters            User ID . : IBMUSER
1  View          Display source data or listings         Time. . . : 07:18
2  Edit          Create or change source data            Terminal. : 3278
3  Utilities     Perform utility functions               Screen. . : 1
4  Foreground    Interactive language processing         Language. : ENGLISH
5  Batch         Submit job for language processing      Appl ID . : ISR
6  Command       Enter TSO commands                      TSO logon : DB13PROC
7  Dialog Test   Perform dialog testing                  TSO prefix: IBMUSER
9  IBM Products  IBM program development products        System ID : VS01
10 SCLM          SW Configuration Library Manager        MVS acct. : ACCT001
11 Workplace     ISPF Object/Action Workplace            Release . : ISPF 8.1
12 z/OS System   z/OS system programmer applications
13 z/OS User     z/OS user applications

     Enter X to Terminate using log/list defaults
```

EXITING FROM THE LAB Z/OS ENVIRONMENT



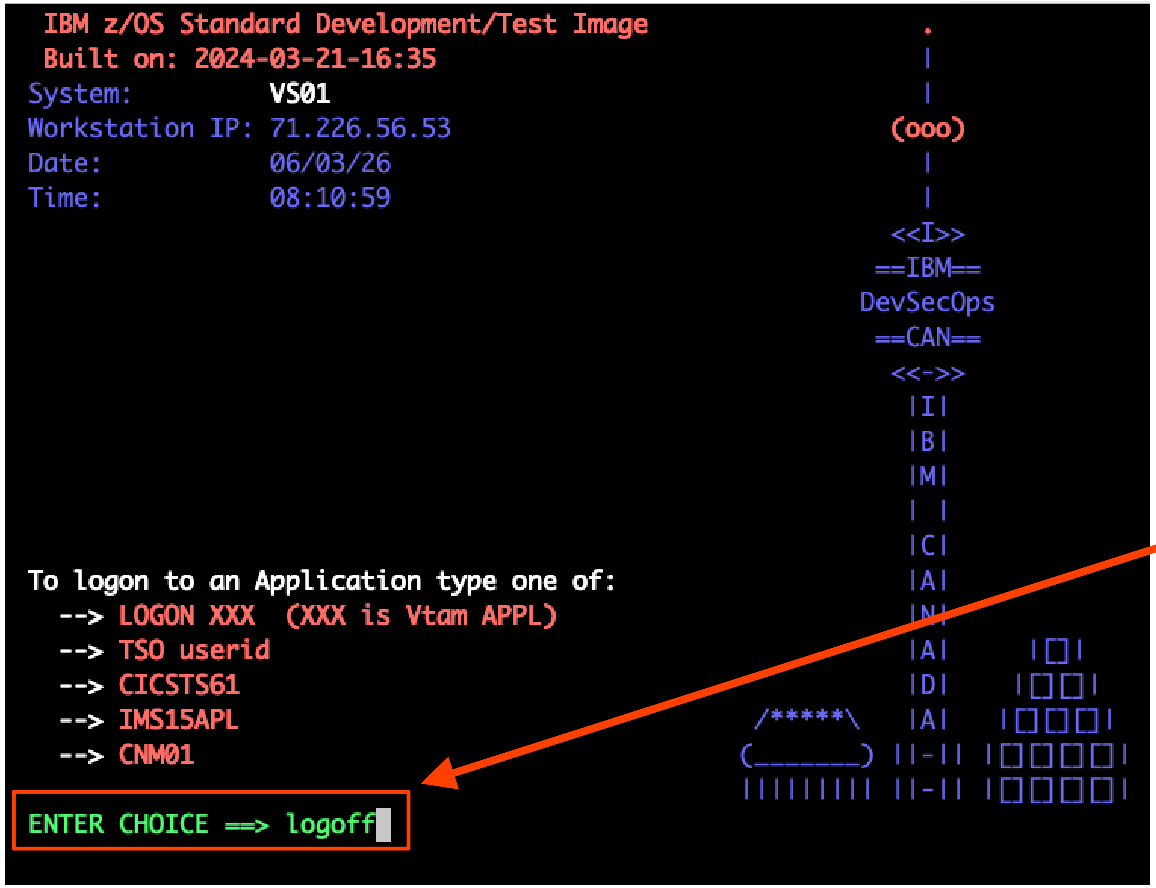
To sign off, type X at the Option prompt and press Enter. That will end ISPF, but not log you off TSO.

EXITING THE LAB Z/OS ENVIRONMENT

Exiting ISPF takes you to TSO. ISPF is a TSO application, and you're still in your TSO session. You can do work on the TSO command line, known as the "ready prompt," or you can log off from TSO.

Log off from TSO by typing "logoff" at the ready prompt and pressing Enter.

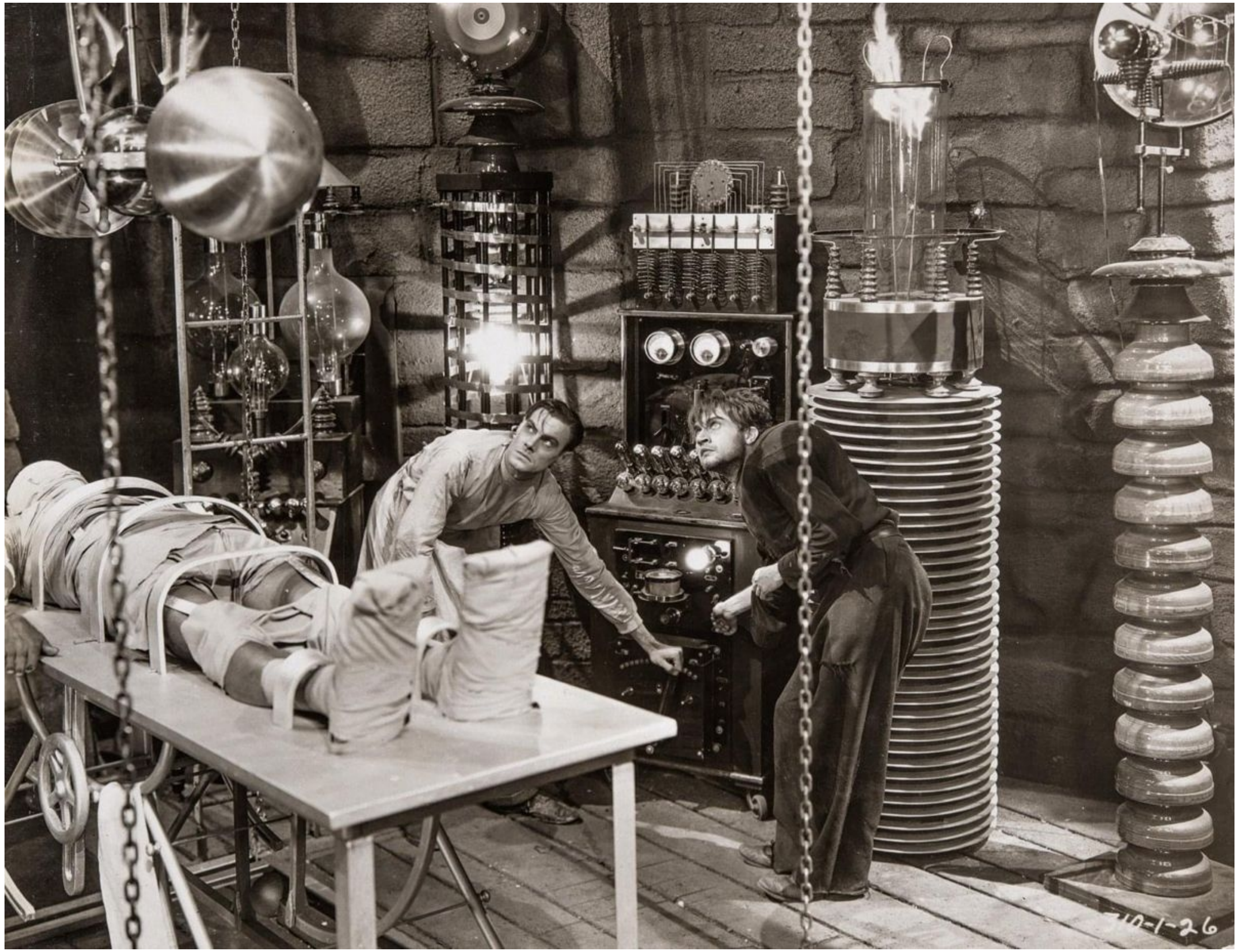
EXITING FROM THE LAB Z/OS ENVIRONMENT



Logging off TSO takes you back to the initial z/OS display.

From here, you can log on to another z/OS system to which you have access, or you can end your terminal emulator session by typing "logoff" and pressing Enter.

LAB Z/OS 2: ACCESS THE Z/OS INSTANCE



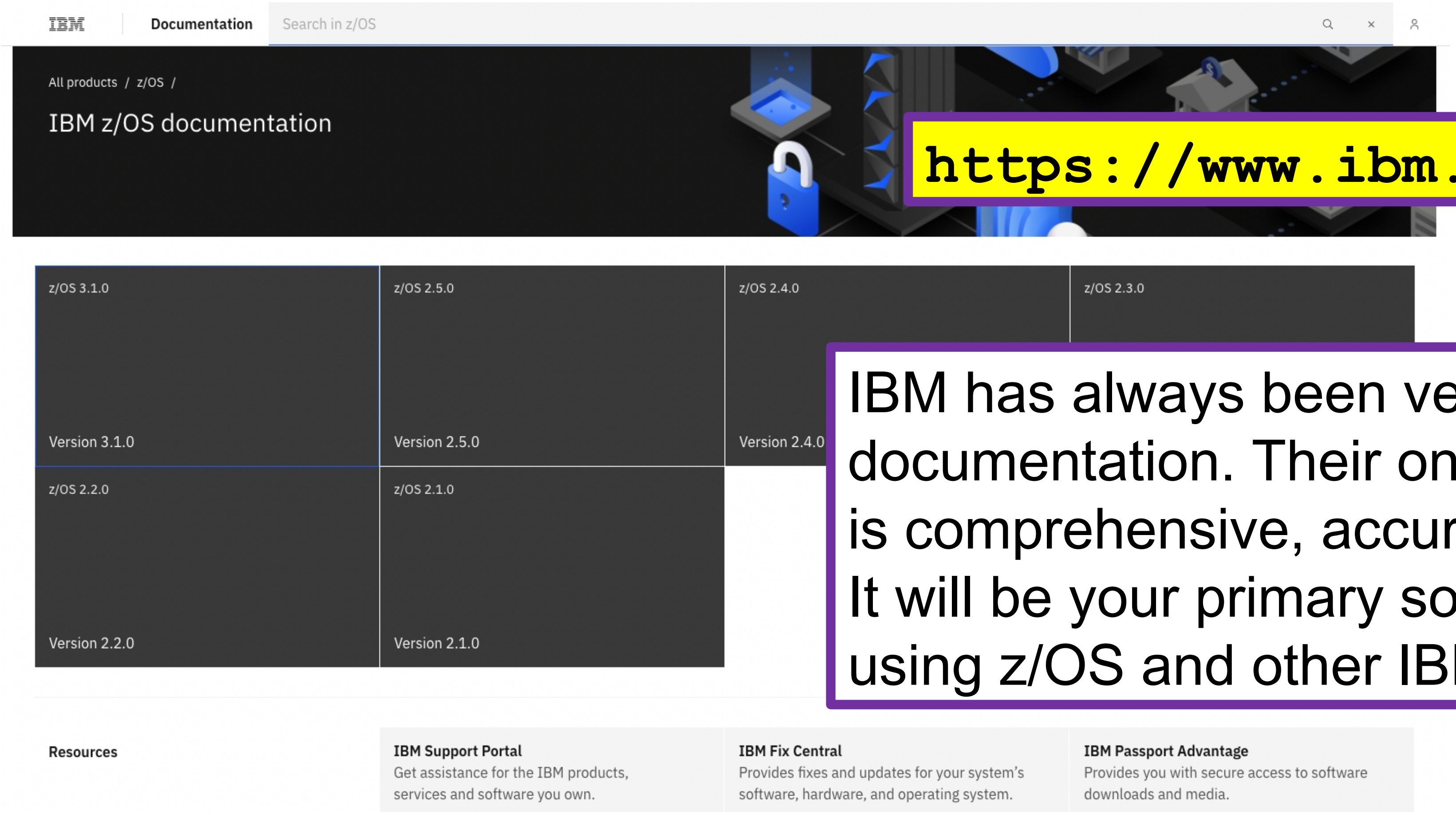
FINDING HELP

Documentation

Community

AI Assist

IBM DOCUMENTATION



https://www.ibm.com/docs/en/zos

IBM has always been very good about documentation. Their online documentation is comprehensive, accurate, and up to date. It will be your primary source of help with using z/OS and other IBM products.

IBM DOCUMENTATION

In the case of IBM products, RTFM (Read the Fine Manual) is sound advice. It is not always given in a respectful or supportive way, but it is sound advice nonetheless.

One of the most useful things you can do is to spend some time exploring the online IBM documentation for z/OS and products you will be using on that platform, such as JCL, SMP/E, and z/OSMF.

Let's see how the documentation is organized and structured, and how to get the most value from it.

IBM DOCUMENTATION – URI STRUCTURE

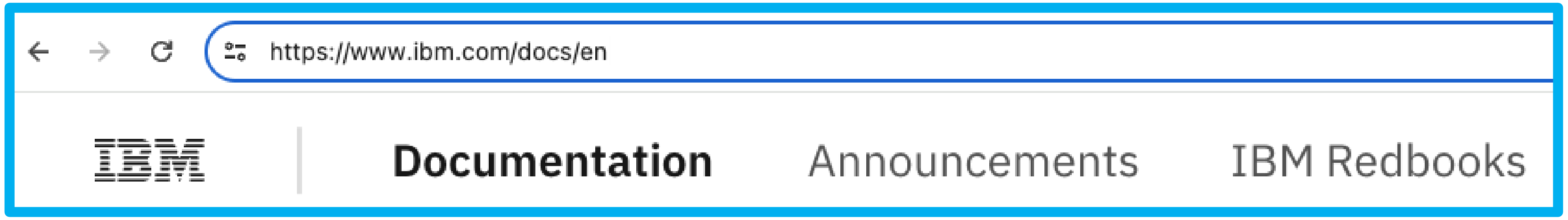
If you're not sure where to start looking for information, you may be able to make a pretty good guess if you understand how IBM structures the URIs for documentation. Here's the top level documentation URI:

`https://ibm.com/docs`

The next segment in the path is the **ISO 639 language code**. The server will default this based on the HTTP Content-Language and/or Accept-Language headers in the request, or on the presumed location of your IP address. In my case, it defaults to English, like this:

`https://ibm.com/docs/en`

Here's the top portion of the resulting document:

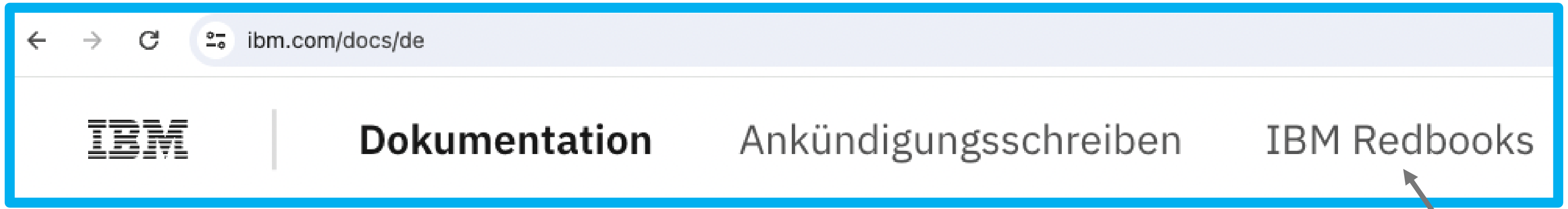


IBM DOCUMENTATION – ISO 639 LANGUAGE CODE

We can change the language code in the URI and reload the page to see the information in other languages, if the particular page is available in the language we specify. Otherwise the default is English.

Here's the same page in German, when we change the language code to **de**.

`https://ibm.com/docs/de`



To get the page in Chinese, we can change the language code to **zh**.

`https://ibm.com/docs/zh`

Trademarked product names are not translated

IBM DOCUMENTATION – PRODUCT NAME

After the language code, the next path segment is a product name. This is not necessarily the official IBM product name, since those often contain slashes or multiple words or a particular combination of upper- and lowercase letters. This will be an abbreviated product name that makes sense as part of a URI. For z/OS, it's **zos** – one of the more guessable examples.

`https://ibm.com/docs/en/zos`

The format of this page is typical of the main page of IBM documentation for a given product. The upper portion contains links to each version of the product for which documentation is available online. IBM rolls older documentation off the Internet as new versions of the product come onto the market. In this case we have the current version of z/OS plus the five most recent previous versions.

IBM DOCUMENTATION – PRODUCT VERSION

The next section of the page contains several links under the heading, **Getting Started**. We'd like to call your attention to one of them, **IBM z/OS Adobe Indexed PDF Collection**. You can obtain nearly all the IBM documentation in the form of PDF files, which you can download free of charge. You may find this format easier to work with for documents you refer to frequently, or that you want to peruse offline.

`https://ibm.com/docs/en/zos/2.5.0`

Getting started	IBM z/OS Adobe Indexed PDF Collection (1.1GB)	What's New in z/OS 2.5.0	z/OS home page
	z/OS 2.5.0 Program Directory	z/OS Introduction and Release Guide	z/OS Release Upgrade Reference Summary

IBM DOCUMENTATION – PRODUCT VERSION

Still farther down the same page, under **Resources**, there are several links to additional documentation and resources.

`https://ibm.com/docs/en/zos/2.5.0`

Resources

IBM Support Portal

Get assistance for the IBM products, services and software you own.

IBM Fix Central

Provides fixes and updates for your system's software, hard operating system.

IBM Z Hot Topics

A favorite source of articles that

Unsurprisingly, most of the in-depth IBM training comes at a price. However, there are quite a few videos and articles available free of charge that provide information at a level of detail suitable for people relatively new to the platform, or to a particular product on the platform.

Redbooks provide "how to" information, experience reports, and configuration guides that usefully augment the reference documentation. They used to be books with red covers. Now they're PDF files and videos.

IBM Redbooks

Find best practices for integrating IBM technologies.

IBM Developer

Explore technical topics, find trial software and join the community.

IBM Training

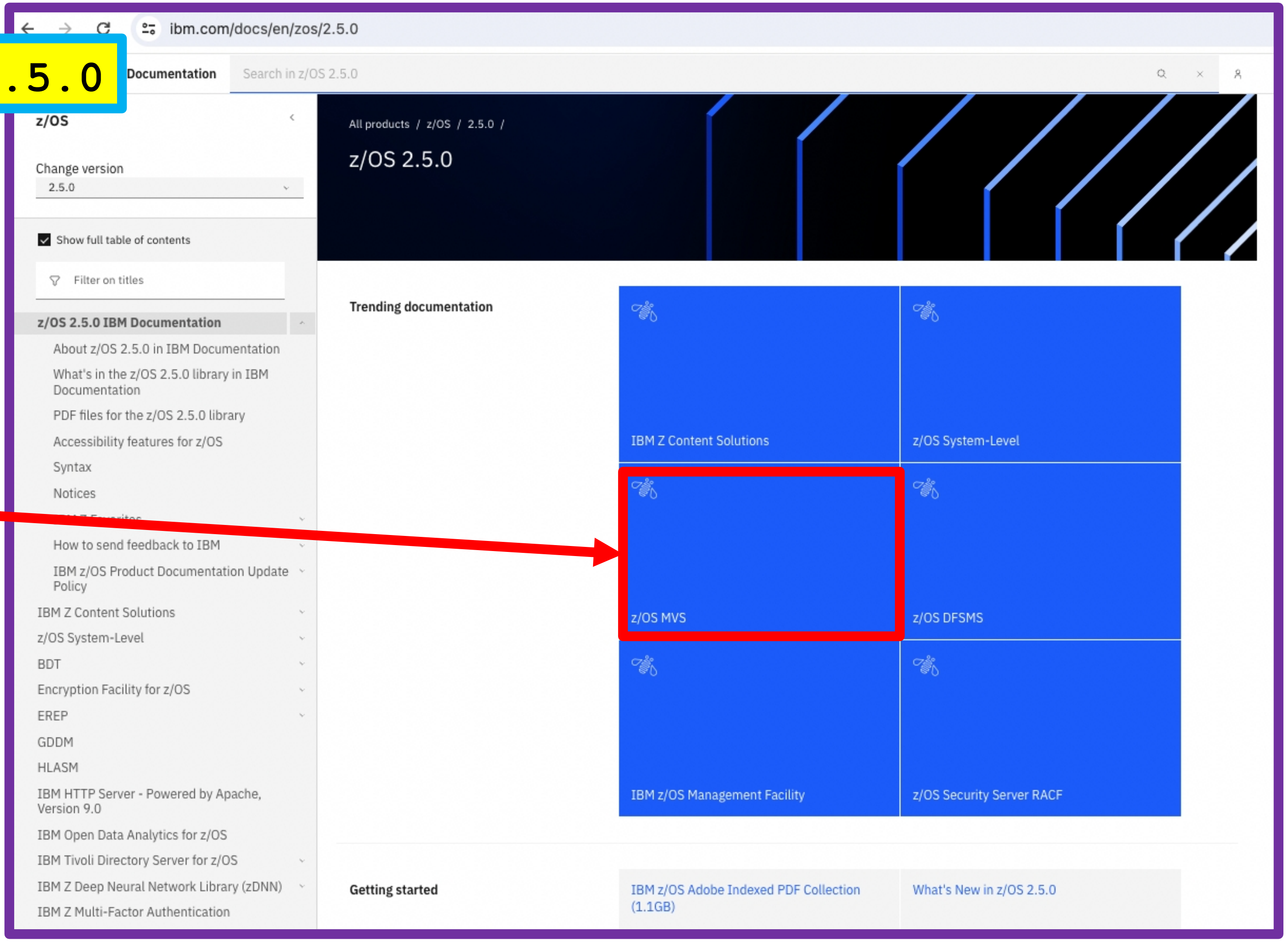
Explore, learn and succeed with IBM training and certification.

IBM Developer is a set of IBM-sponsored and –managed community resources. There are articles, videos, webinars, user groups, and more.

IBM DOCUMENTATION – PRODUCT VERSION

If we follow the link for version 2.5.0, we reach the top documentation page for that version of z/OS.

https://ibm.com/docs/en/zos/2.5.0



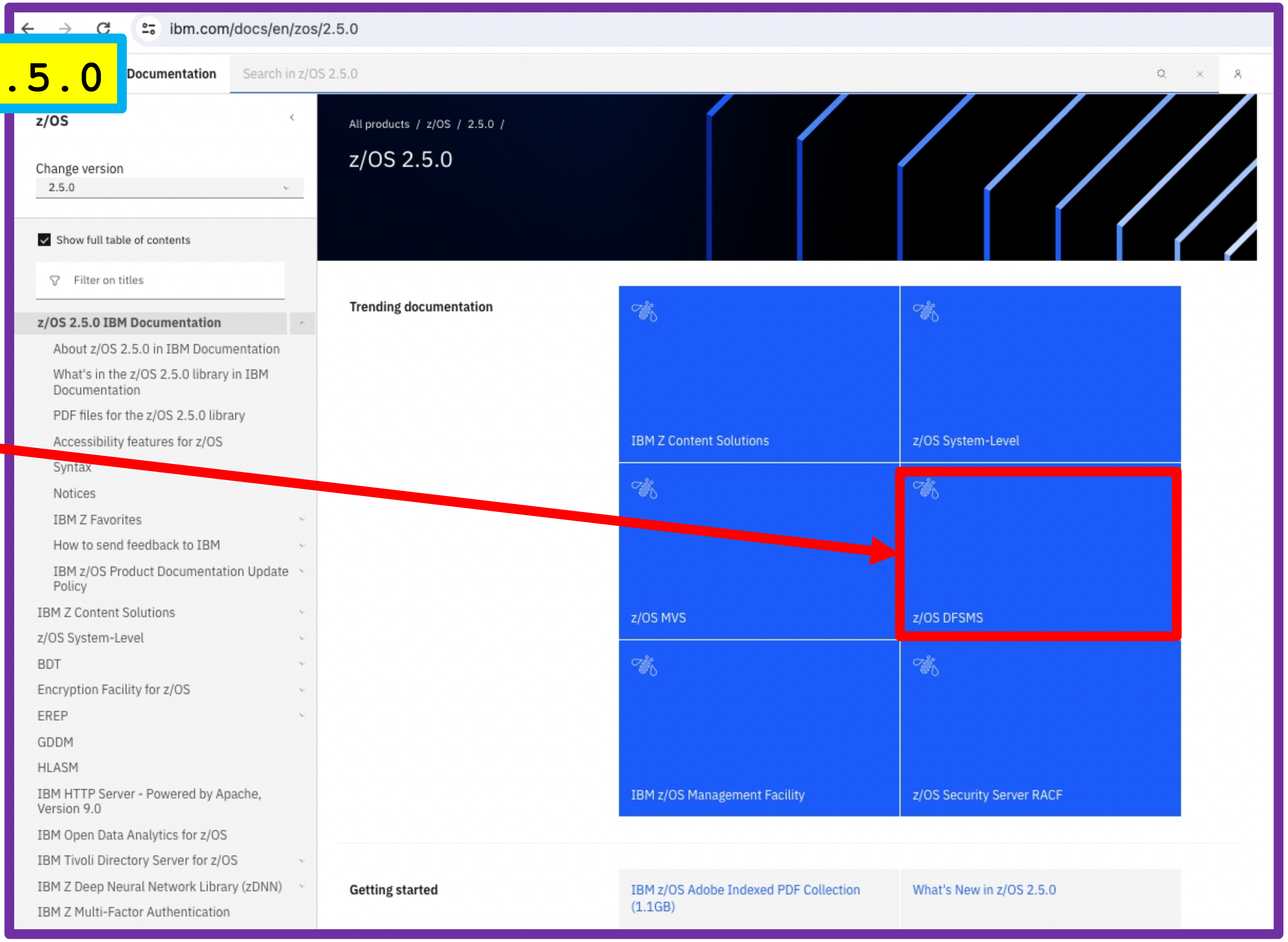
Most of the information we will be interested in for the bootcamp falls under the heading of "z/OS MVS".

MVS (Multiple Virtual Storage) was the name of an ancestor of z/OS. The term MVS refers to elements of z/OS that were part of that operating system. They are still core components of z/OS today.

IBM DOCUMENTATION – PRODUCT VERSION

If we follow the link for version 2.5.0, we reach the top documentation page for that version of z/OS.

https://ibm.com/docs/en/zos/2.5.0



We'll also use the documentation under z/OS DFSMS during the bootcamp.

Let's follow that link now and see where it leads us.

IBM DOCUMENTATION – TOPIC

At this point, the server doesn't add anything to the path, but introduces a query string with the keyword, **topic**.

`https://ibm.com/docs/en/zos/2.5.0?topic=zos-dfsms`

We aren't going to drill into this topic here and now. We want to show you how to navigate the different versions of a topic. All the reference documentation has this sort of navigation.

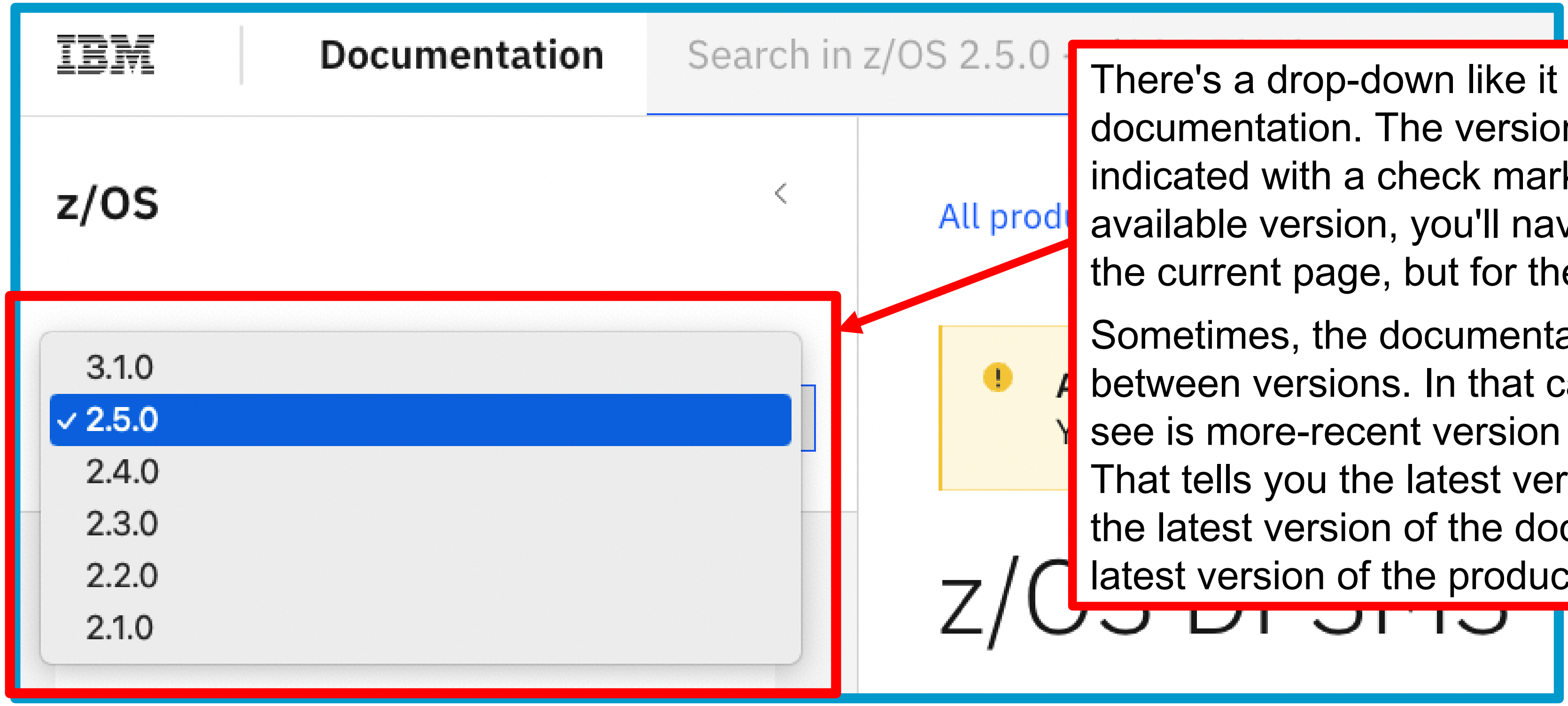
First, the "newer version" pop-up is present because we navigated to this page from a z/OS 2.5.0 page, which is not the most recent version for which online documentation is available. If we follow the **View latest** link, we'll get to the top-level z/OS documentation page for the latest version of z/OS.

But that is not what people usually want. They usually want to go to the latest (or a different) version of the *same topic*. The next slide explains how to do that.

IBM DOCUMENTATION – TOPIC

One way to reach the same topic, but for a different version of the product, is to change the product version in the URI. The other way is to look at the versions listed under the drop-down labeled, "Change version".

https://ibm.com/docs/en/zos/2.5.0?topic=zos-dfsms

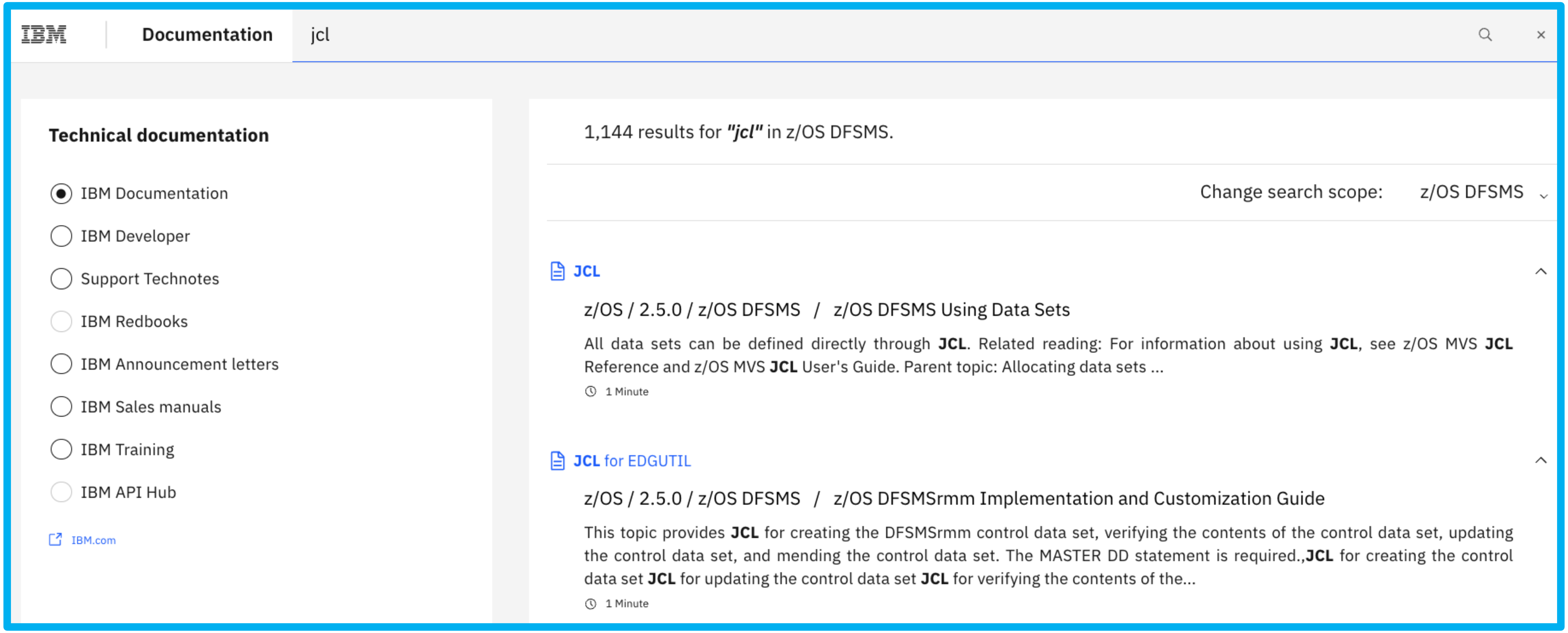


There's a drop-down like it on every page of reference documentation. The version you're looking at is indicated with a check mark. If you select any other available version, you'll navigate to the same topic as the current page, but for the selected version.

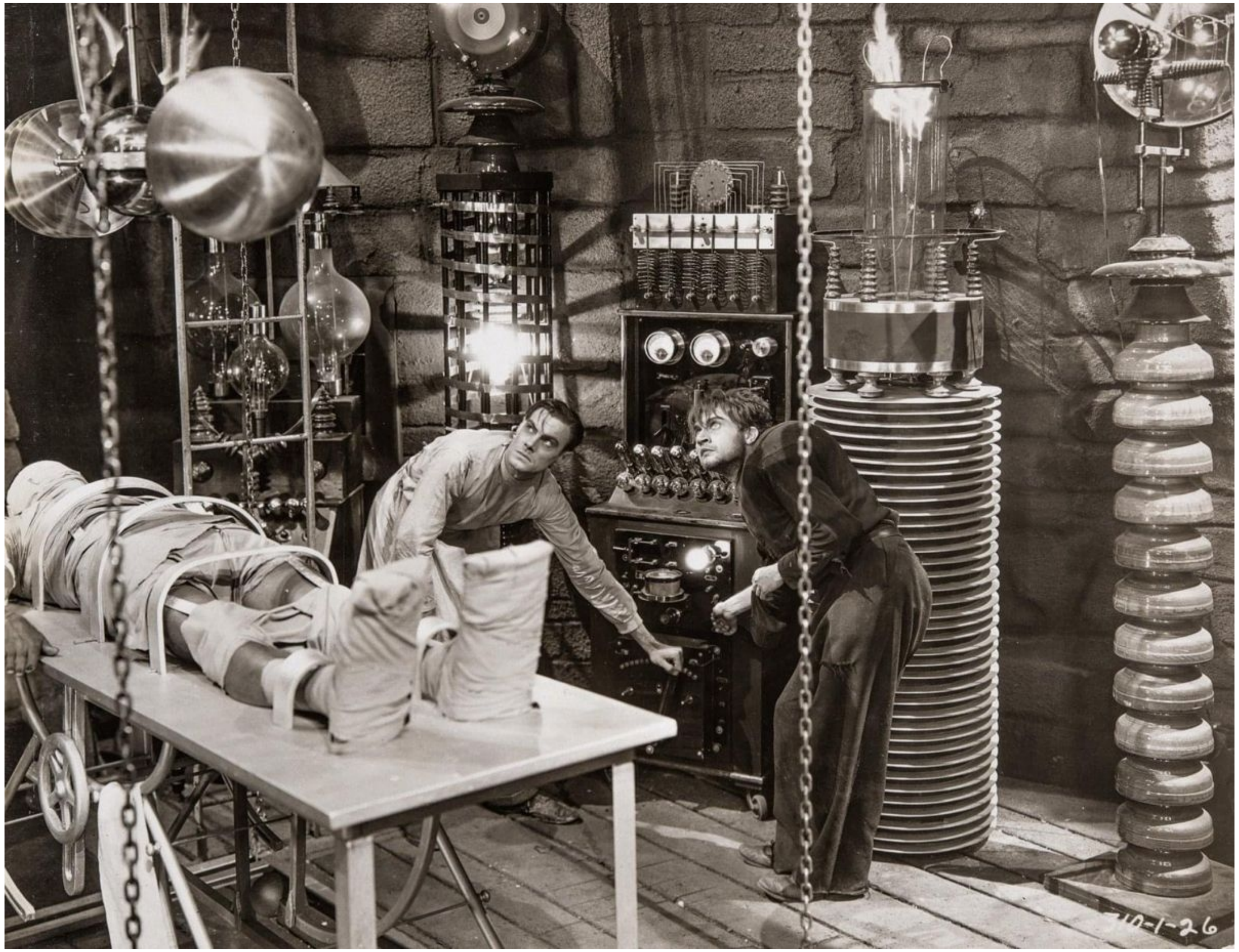
Sometimes, the documentation does not change between versions. In that case, the convention you will see is more-recent version numbers appear in italics. That tells you the latest version in normal typeface is the latest version of the documentation, even if not the latest version of the product itself.

IBM DOCUMENTATION – SEARCHING

There's a search box on every page of reference documentation. If we type in **jcl** and press Enter, we'll get a link to every page of documentation that mentions "jcl". Sometimes we get lucky, but usually we get a very large number of results that are not relevant. It can be quicker to use a generic Internet search engine like DuckDuckGo.



LAB Z/OS 3: EXPLORE IBM DOCUMENTATION



COMMUNITY

There are three kinds of community resources where you might seek help:

- IBM-sponsored and –managed community resources
- Third-party sites that offer explanatory documentation and tutorials
- Discussion boards and mailing lists

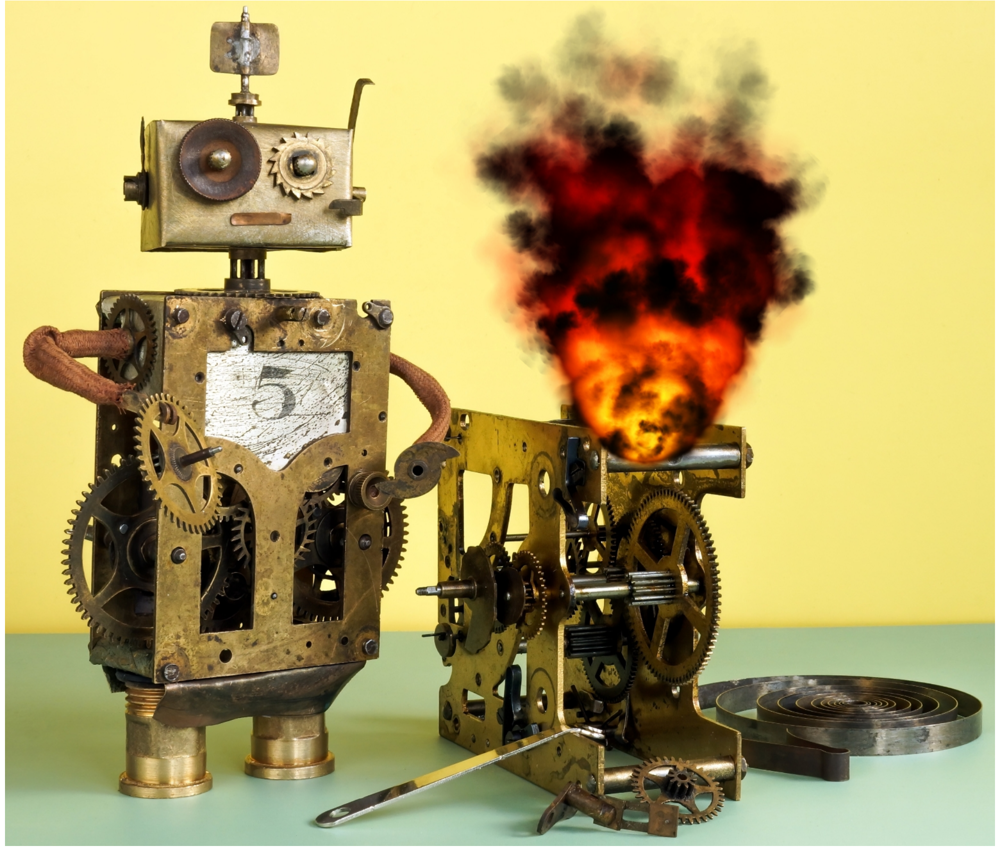
IBM-sponsored resources tend to be open and respectful of newcomers to IBM technologies. The company does not want to alienate new users.

Third-party sites that appear well-organized and present information in an authoritative style are often out of date or just plain wrong in what they say. Exercise caution when using this type of resource.

There are issues with volunteer-run discussion boards focusing on mainframe technologies. **First**, the most prolific people answering questions seem to prefer making others feel stupid over helping them learn. Getting to a clear answer is an unpleasant exercise. **Second**, some of the most helpful are retirees with no access to a live z/OS system. They rely on memory and documentation to answer questions. **Third**, many people describe a workaround that got them through a similar problem in the past, but it is not necessarily a reliable or general solution. They lack true in-depth knowledge of the system. It may be risky to copy what they did.

* maybe

AI ASSIST



Artificial Intelligence tools have become popular in software development. While they may be helpful, it can be risky to depend on them too heavily.

It's useful to remember these tools are not really "intelligent," despite the name.

They are based on Large Language Models (LLMs) trained by examining billions of lines of code accessible on the Internet and in code bases they have been authorized to process.

The general level of quality of software is not so great. AI tools can't make judgments about that. They favor patterns that occur frequently, regardless of quality.

As a result, AI-based coding assistants often provide suggestions that are suboptimal.

The state of the art *circa* 2026 is that knowledgeable programmers can get value from an AI-based assistant because they can reject or modify suggestions the assistant makes.

Novices lack the experience to do that, and they tend to produce questionable results when they use an AI assistant.

This bootcamp is aimed at people who are new to z/OS and related technologies. Therefore, we recommend avoiding AI assistants until you have gained enough experience to understand the quality of the suggestions they offer.

WHAT YOU KNOW AND CAN DO

- You are a committer for your copy of the course repository on Github. You have successfully pushed a text file to the repository that contains your name.
- You have a working IBM 3270 terminal emulation program installed on your local system or accessible to you over the network.
- You have tested your credentials for the z/OS lab environment and you can sign on and off that system.
- You have a general idea of how IBM documentation is organized and structured.

WHAT'S NEXT

- IBM Mainframes